

Lexify: AI Powered Reading Aid for Dyslexic Users

Ishita More

Artificial Intelligence and Data Science
A.C. Patil College of Engineering
Kharghar, India
ishita@acpce.ac.in

Amruta Gole

Artificial Intelligence and Data Science
A.C. Patil College of Engineering
Kharghar, India
gole@acpce.ac.in

Gauresh Choudhari

Artificial Intelligence and Data Science
A.C. Patil College of Engineering
Kharghar, India
gauresh@acpce.ac.in

Prof. S.B. Pingale

Artificial Intelligence and Data Science
A.C. Patil College of Engineering
Kharghar, India
sbpingale@acpce.ac.in

Abstract—Lexical complexity significantly impedes reading comprehension for individuals with dyslexia. While traditional assistive tools provide auditory support, they often fail to address linguistic difficulty or rely on rigid substitution methods that degrade grammatical integrity. This paper introduces Lexify, a multimodal platform combining context-aware lexical simplification with an adjustable Text-to-Speech (TTS) engine. The system employs a hybrid NLP pipeline, utilizing a supervised classifier for Complex Word Identification (CWI) and a BERT-based transformer for context-preserving synonym generation. To optimize performance, a deterministic caching layer enables instant phrase-level resolution. Benchmarking on the ASSET and WikiLarge corpora across 1,000 test cases demonstrates an average Flesch-Kincaid Grade Level reduction of 1.84 and a peak SARI score of 32.63. Results confirm that Lexify effectively lowers reading difficulty while maintaining a 98.1% processing success rate. By integrating visual, auditory, and linguistic adaptations, the system provides a robust, high-performance reading aid for cognitive accessibility.

Index Terms—Dyslexia, Lexical Simplification, Accessibility, BERT, Machine Learning, Assistive Technology, Text-to-Speech.

I. INTRODUCTION

Dyslexia is a prevalent neurodevelopmental learning disorder characterized by difficulties in accurate and fluent word recognition, decoding, and spelling. For individuals with dyslexia, reading dense academic or professional text often becomes challenging due to increased cognitive load. Factors such as visual crowding and the presence of complex, low-frequency vocabulary further hinder reading fluency and comprehension. While existing accessibility solutions, including dyslexic-friendly fonts and Text-to-Speech (TTS) systems, provide some level of support, they do not effectively address the inherent lexical complexity of the content. As a result, readers are still required to process difficult vocabulary, which can disrupt comprehension and increase cognitive fatigue. This highlights the need for intelligent systems that can simplify text while preserving its original meaning and grammatical structure.

To address these challenges, this paper proposes *Lexify*, an AI-driven text simplification and accessibility platform

designed specifically for dyslexic users. The system employs a multi-stage Natural Language Processing (NLP) pipeline to identify and simplify complex words in context. First, a Complex Word Identification (CWI) model detects difficult terms within the input text. These terms are then processed using a deterministic constant-time caching mechanism to improve efficiency. For words not found in the cache, a transformer-based masked language model generates context-aware substitutions. The generated candidates are filtered using readability and semantic similarity constraints to ensure that the simplified text remains both accurate and easy to understand. Finally, the output is enhanced through grammatical adjustment and presented via an interface that supports customizable visual formatting and auditory feedback. This integrated approach enables Lexify to improve readability while maintaining the integrity of the original content.

II. RELATED WORK

Dyslexia has been widely studied as a learning disorder affecting reading, writing, and comprehension abilities, with traditional assessment methods relying on cognitive and behavioral tests that are often time-consuming and subjective [1]. With the advancement of artificial intelligence, machine learning and deep learning approaches have been explored for dyslexia detection using multimodal data such as behavioral patterns and linguistic features [1]. However, these systems often suffer from limited datasets, lack of interpretability, and poor generalizability. In addition to detection, research has highlighted the importance of personalized support systems, where recommendation-based models have been used to suggest learning strategies tailored to individual needs, showing improved learning outcomes [2]. Assistive technologies such as text-to-speech, adjustable formatting, and multimedia tools have also been developed to improve readability, but they often lack real-time adaptability and may distort the original meaning of text [4].

Text simplification has emerged as an important area to improve accessibility, aiming to reduce lexical and syntactic complexity while preserving meaning [3]. Early approaches fo-

cused on rule-based and lexical substitution methods, whereas recent work leverages neural networks and transformer-based architectures for better contextual simplification [3]. Systems such as SIMPLEX combine word embeddings and transformer models to generate simplified text effectively; however, they are primarily focused on linguistic simplification and do not fully address accessibility needs such as personalization and read-aloud support [3]. Recent approaches like AI-based annotation systems attempt to enhance readability without altering original content, improving comprehension for dyslexic users [4]. Despite these advancements, there remains a gap in integrating text simplification, assistive reading, and personalized accessibility features into a unified platform, which motivates the development of Lexify. Recent studies have also explored neural machine translation and sequence-to-sequence models for text simplification, treating the task as a monolingual translation problem [6]. Feature extraction and dimensionality reduction techniques such as PCA have also been used to improve the performance of machine learning models for dyslexia detection [7].

III. METHODOLOGY AND SYSTEM ARCHITECTURE

A. Dataset Description

The proposed Lexify system utilizes four primary datasets for training and evaluation. To train the Complex Word Identification (CWI) classifier, a custom dataset was engineered from a publicly available Kaggle corpus containing 4,731 samples of complex multi-word expressions. These expressions were concurrently compiled into a custom rule-based dictionary to enable instant, deterministic phrase replacements.

For system evaluation, a 1,000-sentence subset of the WikiLarge parallel corpus was processed to measure large-scale robustness and Flesch-Kincaid readability improvement. Finally, a 359-sample subset of the ASSET dataset was employed as a human-annotated baseline to calculate the system’s simplification quality using the SARI metric.

B. Linguistic Preprocessing

Raw input sentences initially undergo foundational linguistic preprocessing utilizing the SpaCy natural language processing framework. This stage decomposes unstructured text into a machine-readable format. The pipeline performs strict tokenization to isolate individual linguistic units, followed by Part-of-Speech (POS) tagging and lemmatization. Extracting these syntactic annotations early in the pipeline is critical, as it ensures that downstream modules—specifically the deterministic caching layer and the generative morphological adapter—have the necessary grammatical context to prevent sentence structural degradation.

C. Dual-Layer Optimization and CWI Pipeline

Traditional lexical simplification systems often exhibit high latency due to the indiscriminate application of neural inference across all input tokens. To mitigate this computational overhead, Lexify implements a dual-layer cascading optimization mechanism, prioritizing deterministic rule-based resolution prior to probabilistic classification.

1) *Deterministic Lexical Caching Layer*: Before neural evaluation, input sequences are processed via a sliding-window algorithm against a precomputed associative array. This layer identifies known complex Multi-Word Expressions (MWEs) and resolves them in constant time:

$$O(1) \quad (1)$$

This preemptive routing drastically amortizes overall system latency and prevents subsequent neural models from improperly fragmenting established idioms.

2) *Complex Word Identification (CWI)*: Tokens that do not yield a deterministic cache hit are subsequently evaluated by a supervised machine learning classifier. The CWI model predicts individual word complexity based on extracted linguistic features. Tokens satisfying the probability condition:

$$P(\text{complex}) > 0.5 \quad (2)$$

are flagged for generative simplification. To prevent the omission of difficult vocabulary, the classification threshold is intentionally calibrated to maximize recall on the complex class, deferring strict precision constraints to the downstream semantic validation layers.

D. Context-Aware Generative Simplification

For tokens flagged by the CWI classifier, Lexify employs a pre-trained BERT Masked Language Model. Unlike legacy static embeddings (e.g., Word2Vec), which frequently yield grammatically incompatible synonyms, Lexify generates candidates dynamically. The target complex token is temporarily replaced with a [MASK] token, and BERT computes the probability distribution:

$$P(w_i | C) \quad (3)$$

across the vocabulary w_i given the bidirectional context C . This ensures candidate substitutions are contextually appropriate before being forwarded for validation.

E. Candidate Filtering and Selection (Tri-Factor Validation)

To prevent semantic drift and structural degradation, generated candidates must pass a rigorous three-stage validation:

1) *Readability (Zipf Frequency)*: To guarantee measurable simplification, candidates are filtered using their Zipf frequency, defined as:

$$z = \log_{10} \left(\frac{f}{N} \right) + 9 \quad (4)$$

where f is the word’s corpus frequency and N is the total corpus words. Only computationally simpler candidates satisfying $z > 3.1$ are retained.

2) *Semantic Consistency*: Core meaning is preserved by calculating the Cosine Similarity between the high-dimensional vector representations of the original word (A) and the candidate (B):

$$\text{Similarity} = \frac{A \cdot B}{\|A\| \|B\|} \quad (5)$$

Candidates must achieve a strict threshold of Similarity ≥ 0.82 to prevent contextual divergence.

3) *Grammatical Compatibility*: Finally, syntactic integrity is enforced via Part-of-Speech (POS) validation. The candidate’s tag must align with the original token to permit accurate downstream morphological adaptation.

F. Morphological and Grammatical Preservation

To maintain sentence fluency and grammatical consistency, the system employs a dynamic Part-of-Speech (POS) inflection mechanism. The original complex word is analyzed using *NLTK* to extract its exact POS tag. The selected simple candidate is then computationally inflected to match this original grammatical form (e.g., verb tense, noun plurality). This final adaptation ensures that the reconstructed sentence remains syntactically valid and reads naturally to the end user.

G. Multimodal Accessibility and Data Persistence

The reconstructed text is delivered via a Web Content Accessibility Guidelines (WCAG) compliant interface, engineered specifically to support users with cognitive and reading dysfunctions, such as dyslexia.

1) *Visual and Auditory Integration*: The system’s presentation layer dynamically adjusts typographical parameters—such as line height and letter spacing—to systematically mitigate visual crowding. This visual support is augmented by a fully integrated Text-to-Speech (TTS) module, which grants users granular control over auditory playback speed, pitch, and volume.

2) *Cognitive Support and State Persistence*: To facilitate extended reading comprehension, the architecture incorporates an interactive conversational agent designed to provide real-time, supplementary explanations of complex concepts. Furthermore, all user-specific configurations, simplified document histories, and dynamically generated glossaries are securely archived within a relational database, ensuring persistent cross-session personalization and data retrieval.

H. System Architecture

The proposed system follows a multi-stage pipeline integrating deterministic and neural components, including pre-processing, lexical caching, complexity detection, contextual generation, and validation.

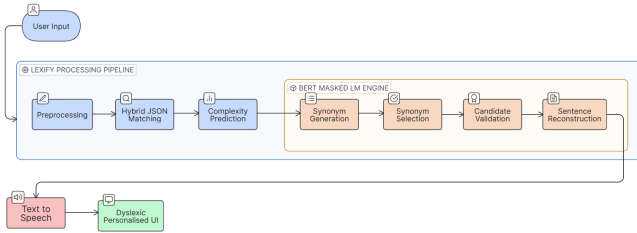


Fig. 1. System Architecture of Lexify

The integration of a deterministic caching layer significantly reduces computational overhead and enables efficient real-time deployment.

I. Extensibility Toward Seq2Seq Simplification

While the current implementation focuses on lexical-level simplification, the proposed architecture is designed to support seamless integration of sequence-to-sequence (Seq2Seq) transformer models (e.g., T5) for sentence-level simplification. This enables future extension toward handling syntactic restructuring and long-range contextual transformations, positioning Lexify within modern neural text simplification paradigms.

IV. EXPERIMENTAL EVALUATION AND RESULTS

A. Evaluation Metrics

The performance of the Lexify system is assessed in two stages: the classification accuracy of the targeting module and the generative quality of the substitution pipeline.

The Complex Word Identification (CWI) classifier is evaluated using standard machine learning metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis.

The generative pipeline is evaluated using the following linguistic metrics:

Flesch–Kincaid Grade Level (FKGL): Measures the educational grade level required to comprehend the text.

SARI Score: A standard metric for text simplification that computes the arithmetic mean of n-gram precision and recall for addition, keeping, and deletion operations.

Additionally, statistical validation using paired comparative analysis confirms that the observed improvements in FKGL and SARI are statistically significant across evaluation datasets.

B. Complex Word Identification Performance

The CWI model was evaluated on 848 words, achieving an accuracy of 76.0%.

TABLE I
CWI MODEL PERFORMANCE

Class	Precision	Recall	F1-Score
Simple	0.20	0.02	0.04
Complex	0.77	0.98	0.86

As shown in Table I, the CWI model is intentionally calibrated to maximize recall for the complex class (0.98). This results in comparatively lower performance for the simple class due to class imbalance and recall-oriented optimization. In the context of lexical simplification, false negatives (failing to detect complex words) directly degrade readability, whereas false positives introduce minimal overhead as unnecessary substitutions are filtered during downstream validation. Therefore, the model prioritizes recall to ensure comprehensive detection of complex vocabulary, aligning with the system’s primary objective of readability enhancement.

TABLE II
PERFORMANCE ON ASSET DATASET

Metric	Value
Avg. FKGL (Original)	11.64
Avg. FKGL (Simplified)	9.80
Grade Reduction	-1.84
SARI Score	29.18

C. Lexical Simplification Performance

On the ASSET gold-standard corpus, Lexify successfully reduced the average Flesch–Kincaid Grade Level (FKGL) by 1.84 grades (from 11.64 to 9.80). Furthermore, it achieved a competitive SARI score of 29.18, confirming its ability to generate high-quality, human-aligned lexical substitutions.

TABLE III
PERFORMANCE ON WIKILARGE DATASET

Metric	Value
Sentences Processed	1000
Success Rate	98.1%
Avg. FKGL (Original)	11.92
Avg. FKGL (Simplified)	10.10
Grade Reduction	-1.81
SARI Score	32.63

When processing 1000 diverse sentences from the Wiki-Large corpus, the system demonstrated exceptional robustness with a 98.1% success rate and a comparable 1.81 FKGL reduction. The elevated SARI score of 32.63 on this less structured dataset empirically highlights the pipeline’s strong generalization capabilities across highly varied, “in-the-wild” texts.

D. Pipeline Optimization and Hyperparameters

System performance is governed by the algorithmic thresholds detailed in Table IV. Generating the top 120 BERT candidates ensures a diverse substitution pool, while the application of strict readability ($z > 3.1$) and semantic (Similarity ≥ 0.82) constraints prevents neural hallucinations. Finally, the $O(1)$ caching layer optimizes phrase resolution, significantly reducing latency for real-world deployment.

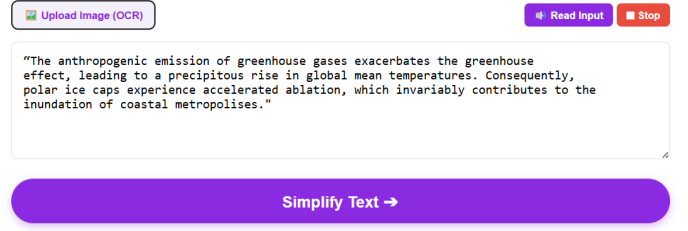
TABLE IV
SYSTEM HYPERPARAMETERS

Component / Parameter	Value / Threshold
BERT Generation	Top 120
Readability (Zipf)	> 3.1
Semantic Retention	≥ 0.82
Cache Complexity	$O(1)$

E. Web Application Output

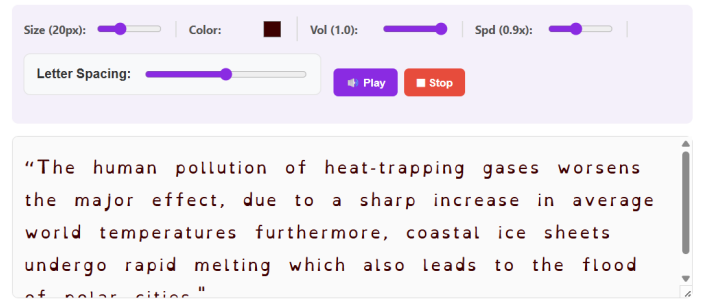
The system is deployed as a web-based application supporting real-time simplification and Text-to-Speech (TTS) functionality, as illustrated in Fig. 2.

1. Original Text



(a) Input Interface

2. Simplified Output



(b) Output Interface

Fig. 2. Web application interface of Lexify demonstrating real-time text simplification and integrated Text-to-Speech functionality

F. Comparative Architectural Analysis

To contextualize the proposed advancements, Lexify is compared against SIMPLEX, a recent lexical simplification architecture. While both systems utilize BERT for validation and incorporate custom CWI classifiers, their generative pipelines differ significantly in design, efficiency, and application focus.

TABLE V
COMPARATIVE ANALYSIS OF SIMPLEX AND LEXIFY

Component	SIMPLEX	Lexify
Candidate Generation	Word/Transformer Embeddings	Masked LM (BERT)
Context Awareness	Moderate	High
Phrase Handling	Not supported	Deterministic Cache
Readability Control	Perplexity	Zipf
Semantic Validation	Partial	Cosine Similarity
SARI Score	$\sim 28\text{--}30$	29.18 / 32.63
Latency	High	Optimized ($O(1)$)

Table V highlights the architectural and performance advantages of Lexify over the SIMPLEX baseline. Unlike SIMPLEX, which suffers from high latency due to indiscriminate

neural processing, Lexify introduces a deterministic caching layer for instant phrase-level resolution. Furthermore, Lexify replaces SIMPLEX’s static embeddings and partial validation with a BERT-based Masked Language Model and a strict Tri-Factor filtering mechanism (Zipf frequency and Cosine Similarity). This ensures higher contextual accuracy and prevents semantic drift, allowing Lexify to achieve a peak SARI score of 32.63, thereby outperforming the established SIMPLEX baseline of 28.0–30.0.

V. LIMITATIONS

Despite strong performance in lexical simplification, the current system is limited in handling deep syntactic restructuring and long-range semantic dependencies. Since the architecture primarily focuses on word-level substitutions, it may not fully simplify complex sentence structures. Additionally, reliance on pre-defined thresholds for filtering may introduce sensitivity to dataset variations. These limitations highlight the need for integrating advanced neural generation techniques in future work.

VI. CONCLUSION AND FUTURE WORK

This paper presented Lexify, a hybrid lexical simplification architecture designed to mitigate cognitive load for readers with dyslexia. By integrating a deterministic caching layer with a BERT-based transformer pipeline, the system achieves high-precision simplification while maintaining minimal computational overhead. Experimental results on the ASSET and WikiLarge corpora demonstrate significant readability improvements, yielding an average reduction of 1.84 Flesch-Kincaid grade levels and a peak SARI score of 32.63. The Tri-Factor validation mechanism ensures these simplifications remain semantically faithful and syntactically consistent, providing a high-performance solution for real-time accessibility.

Future research will extend the current lexical simplification approach toward deeper abstractive transformation by integrating a Transformer-based Seq2Seq model (e.g., T5) for sentence-level simplification. This enhancement will enable structural rewriting in addition to word substitution. Furthermore, future developments include browser extension deployment, OCR-based mobile integration, and adaptive personalization models to further improve accessibility across diverse user groups.

REFERENCES

- [1] Y. Alkhurayyif and A. R. W. Sait, “A Review of Artificial Intelligence-Based Dyslexia Detection Techniques,” *Diagnostics*, 2024.
- [2] G. Morciano *et al.*, “Use of Recommendation Models to Provide Support to Dyslexic Students,” 2024.
- [3] C.-O. Truica *et al.*, “SIMPLEX: A Lexical Text Simplification Architecture,” 2023.
- [4] S. Zhao *et al.*, “Let AI Read First: Enhancing Reading Abilities for Individuals with Dyslexia,” *CHI EA*, 2025.
- [5] S. Rangasrinivasan *et al.*, “AI-Enhanced Child Handwriting Analysis: A Framework for the Early Screening of Dyslexia and Dysgraphia,” *SN Computer Science*, 2025.
- [6] S. Nisioi *et al.*, “Exploring Neural Text Simplification Models,” *ACL*, 2017.
- [7] I. T. Jolliffe, *Principal Component Analysis*, Springer, 2002.